1 Introduction to Software Engineering Management

Week 1 · Cheat sheet



Set up once

`git config --global user.name "Your Name"`	Name stored in every commit
`git config --global user.email "you@example.com"`	Email stored in every commit
`git init -b main`	Start a repository with main as the branch

Daily loop

`git status`	What changed, what is staged
`git diff`	Unstaged changes
`git add <file>`	Stage a file
`git diff --staged`	What the next commit will contain
`git commit -m "add sum"`	Record staged changes (imperative message)
`git restore --staged <file>`	Unstage without losing edits

Reading history

`git log --oneline`	One commit per line
`git log --oneline --graph --all`	Branch structure of every branch
`git show <hash>`	Message and diff of one commit

Branches and merges

`git branch`	List branches (* = current)
`git switch -c feature-x`	Create and switch to a branch
`git switch main`	Switch to an existing branch
`git merge feature-x`	Merge feature-x into the current branch

Remotes	
`git remote add origin <url>`	Name a remote "origin"
`git push -u origin main`	Upload main and set upstream
`git fetch origin`	Download commits; your files do not change
`git pull origin main`	Fetch and integrate

When something goes wrong

Symptom	Likely cause	Fix
Author identity unknown	Name or email not configured	git config --global user.name / user.email
Conflict markers in a file	A merge stopped	Edit, remove markers, git add, git commit
Push rejected (non-fast-forward)	The remote has commits you lack	git fetch, integrate, push again. Never --force on shared branches.